

Lab Worksheet R01 — Intro to Python: Variables, Types, Operations & Expressions

Use: "Recitation-style Lab B" segment (40 min) of a 3-hour session. **Ground rule:** The TA never gives away answers — the TA only asks tracing questions. **Source note:** Recitation 1 contains no separate practice problems; every exercise below is rearranged from the single example-code slide in the original notes.

Warm-up (about 5 min)

The opening lines of the original example code. Fill in the blanks to complete it.

```
# declaring a simple string
basic_string = _____      # (1) a sentence in double quotes, e.g. with
                                your name

# integers and simple operations
a = 1
b = _____                # (2) assign the integer 20
sum_ab = a _____ b      # (3) the sum of the two numbers
```

After completing it, run `type(basic_string)` and `type(a)` in the console and write down what each returns: _____ / _____

Core exercises (about 30 min)

Exercise 1. Integers and simple operations

(a) **Worked example** — read and run this original code as-is.

```
a = 1
b = 20
sum_ab = a + b      # sum_ab is now 21
difference_ab = b - a  # difference_ab is now 19
```

(b) **One-line change** — change only the first line to `a = 5`. Before running, predict the new values:

- `sum_ab = _____`, `difference_ab = _____`

(c) **Fill in the blank** — complete the code so it computes the same two results.

```
a = 5
b = 20
sum_ab = _____ + _____
difference_ab = _____ - _____
```

[TA tracing questions]

1. "Right after line 3 runs, what value is stored in `sum_ab`? And in `difference_ab` — does it exist yet?"
2. "Why do `b - a` and `a - b` give different results? What would you change to get -19?"

Exercise 2. `str()` conversion and string concatenation

(a) **Worked example** — this line is from the original slide. Run it, then print the variable.

```
# example operations on integers and strings
my_numbers_as_string = str(9) + ' ' + str(10) + ' ' + str(7)
```

(b) **One-line change** — replace `str(7)` with `str(a)` (with `a = 5` from Exercise 1). Predict the printed result before running: _____

(c) **Fill in the blank** — complete the line so it produces exactly `"9 10 7"`.

```
my_numbers_as_string = ____ (9) + ' ' ____ str(10) + ____ + str(7)
```

[TA tracing questions]

1. "What is the difference between `9` and `str(9)`? What does `type()` say for each?"
2. "If you remove `str()` and write `9 + ' '`, what happens — and why does Python refuse?"

Self-check (about 5 min)

Check each box only if you can do it **without looking at the completed code**.

- ☐ I can declare a string variable and an integer variable, and explain how their types differ.
- ☐ I can predict the values of `sum_ab` and `difference_ab` for any pair of `a`, `b` before running.
- ☐ I can explain why `str()` is needed before joining a number to a string with `+`.

Source

- MIT OpenCourseWare, **6.100L Introduction to Computer Science and Programming Using Python** (Fall 2022), **Recitation 1** notes.
- License: **CC BY-NC-SA 4.0**. This worksheet is a scaffolded rearrangement of the original example code; no new problems were invented.
- <https://ocw.mit.edu> — see the course page for citation and Terms of Use.